

Yassine Ghorbel

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AI Engineer • Agents, Retrieval & Evaluation • Tools for non-technical teams

Profile

AI engineer who builds **AI agents** that carry out multi-step tasks, **search systems** that find answers in large collections of documents, and the **tests** that prove they work, so that non-engineers can trust the results. I understand how large language models (LLMs) behave in practice: how much text they can read at once, how to connect them to documents and tools, and where **fine-tuning** helps and where it does not. I write **prompts**, plug AI into existing software, and work with **Claude**, Cursor and Codex. I pick the model that scores best on our tests, not simply the biggest one, and I teach my methods so they do not depend on one person. Main tools: LangGraph, LangChain, graph-based RAG, FastAPI. Based in Tunis.

Core Competencies

Building AI agents • Document search (RAG) • Testing before claiming results • Decisions backed by evidence • Tools for non-engineers • Teaching in plain language • Working with clients • Taking research to production

Experience

Guepard | *AI Intern: NL2SQL, asking databases questions in plain English* — Feb 2026 – Jul 2026

Full-time, Remote · Paris, France

- Built an 8-step **LangGraph** pipeline that turns a Postgres database into a structured map of its tables and meaning (an OWL/SKOS ontology). It also infers foreign-key links from the data itself, recovering ∼**37%** of the benchmark's table joins that the database never declares.
- Trained a **retriever** (bge-small-en-v1.5, PyTorch, InfoNCE contrastive learning) that matches questions to the right tables. I split training and test data by query pattern, so the model is never tested on a pattern it has already seen.
- About one third of the correct tables were "bridge" tables that similarity search cannot find. Instead of hiding this weakness, I redesigned retrieval (Steiner-tree search with gating) to handle it.
- Integrated this into Qwery's LangGraph SQL agent, with schema validation and **LangSmith** tracing. Accuracy on BIRD rose from **32.5% to 59%** and on Mini-Dev from **29% to 66.4%**, while token use fell to ∼**0.38x**. I optimized accuracy first, then cost.

Allianz Technology | *Research Intern: Insurance Policy Design* — Oct 2025 – Jan 2026

Part-time, Remote · Munich, Germany

- Scanned Tunisian and EU insurance regulations (OCR) and built a **Neo4j** knowledge graph of **1.9K nodes**, so compliance questions are answered by following links between rules rather than guessing from keywords.
- Set up an **evaluation** with **30 labeled questions**. Adding each result's neighbouring graph nodes raised recall@10 (the share of correct answers found in the top 10) from **0.62 to 0.77**.
- Weighed whether **fine-tuning** was worth it under cost and on-premise constraints. I chose a LangChain planner with **85%** accuracy and a self-hosted 2.6B-parameter model (LFM2), after it beat six larger models under 4B parameters.
- Made it usable: everything runs locally (MinIO, Qdrant, Neo4j) and can move to AWS, with React/Next.js/Apollo screens for non-technical stakeholders.

Guepard | *Community Builder* — Mar 2025 – Present

Part-time, Remote

- Built React/Next.js/Apollo GraphQL features that users actually touch, alongside running the community.
- Made tutorial videos and blog posts so new users can get started without asking someone. Worked directly with clients (Linkup, Monderal, Visum) and fed their feedback back to the team.

Huawei R&D | *AI Intern: Network Issue Classification with MLOps* — Jul 2025 – Aug 2025

Full-time, On-site · Tunis, Tunisia

- Cleaned messy network data: **5,525** records of cell-tower (eNodeB) performance indicators, cut from 107 to 52 columns. Added time-based features, labeled the data with K-Means clustering, and balanced the classes with SMOTE.
- Delivered a tool the team uses daily: a multi-label **FastAPI** service handling **10,000+ records per second in under 100 ms**, with automated pipelines (Prefect, MLflow, Docker CI/CD).

CRNS (Digital Research Center of Sfax) | *AI Intern: Brain Tumor Classification* — Jun 2024 – Aug 2024

Full-time, On-site · Sfax, Tunisia

- Closed a 26-point gap between validation and test accuracy, reaching **98.2%** test accuracy. Built a DCGAN (image-generating model), then recommended against using it after finding it relied on shortcuts in the images rather than real signs of tumors.
- Deployed both models behind a **Gradio** API with Swagger documentation, so the team can try them in a web browser instead of code notebooks.

Projects

- **Spatial Reasoning AI Agent** (SUP'COM): an agent that uses tools plus a vision-language model (VLM) to answer questions about the layout of a warehouse; runs on AWS.
- **Insurance Feature Intelligence Platform** (SUP'COM × Allianz): analyzed 650K+ policies; reduced prediction error (Tweedie deviance) by **2.1%**; built 25+ data-quality checks that catch **94%** of anomalies.
- **IndabaX 2025: 1st Place**. **TSYP 13** (Orange/IEEE): Technical Lead and Jury member; mentored participants on AWS.

Education

ICT Engineering Degree | *Higher School of Communications Tunis (SUP'COM)* — Sept 2023 – Sept 2026

AI Agents for Services, MLOps, Core CS; graduated Sept 2026

Ranked 164/1500 | *Faculty of Sciences of Sfax* — Sept 2021 – Aug 2023

Advanced Mathematics, Theoretical Physics, Python

Skills

Programming	Python, TypeScript, SQL, C++, Rust, R, MATLAB
AI / LLMs	LangGraph, LangChain, LangSmith, RAG, LLMs, Claude, Cursor, Codex, prompts, context windows, fine-tuning limits, tool use, integrations, PyTorch, Gradio
Search & Data	Neo4j, Qdrant, MinIO, PySpark, Delta Lake, Kafka
Production & Web	FastAPI, Docker, CI/CD, Prefect, MLflow, React, Next.js, Apollo GraphQL, AWS

Languages

English (Fluent, TOEIC 850) • French (B2, diploma-certified) • German (A2) • Spanish (A2)